# Chapter 4. Introduction to Generalized Linear Models

*Exponential family, likelihood, deviance, fitting*

## Contents


**4.1 Generalized linear model** **2**
- 4.1.1 Components of generalized linear models . . . 2
- 4.1.2 Binomial logit models for binary data . . . 2
- 4.1.3 Poisson loglinear models for count data . . . 2
- 4.1.4 Generalized linear models for continuous responses . . . 2
- 4.1.5 Deviance . . . 3
- 4.1.6 Advantages of the GLM formulation . . . 3

**4.2 Generalized linear models for binary data** **3**
- 4.2.1 Linear probability model . . . 4
- 4.2.2 Snoring and heart disease example . . . 4
- 4.2.3 Logistic regression model . . . 4
- 4.2.4 Binomial GLM for $2 \times 2$ contingency tables . . . 5
- 4.2.5 Probit and inverse cdf link functions . . . 5

**4.3 Generalized linear models for counts** **5**
- 4.3.1 Poisson loglinear models . . . 5
- 4.3.2 Horseshoe crab mating example . . . 5
- 4.3.3 Overdispersion for Poisson GLMs . . . 6
- 4.3.4 Negative binomial GLMs . . . 6
- 4.3.5 Poisson regression for rates . . . 7
- 4.3.6 Poisson GLM of independence in $I \times J$ contingency tables . . . 7

**4.4 Moments and likelihood for generalized linear models** **7**
- 4.4.1 Mean and variance functions for the random component . . . 7
- 4.4.2 Mean and variance functions for Poisson and binomial . . . 8
- 4.4.3 Systematic component and link function . . . 8
- 4.4.4 Likelihood equations for a GLM . . . 8
- 4.4.5 Likelihood equations for binomial GLMs . . . 9
- 4.4.6 Asymptotic covariance matrix of model parameter estimators . . . 9
- 4.4.7 Likelihood equations and covariance matrix for Poisson loglinear model . . . 10

**4.5 Inference for generalized linear models** **11**
- 4.5.1 Deviance and goodness of fit . . . 11
- 4.5.2 Deviance for Poisson models . . . 12
- 4.5.3 Deviance for binomial models: grouped and ungrouped data . . . 12
- 4.5.4 Likelihood-ratio model comparison using the deviance . . . 13
- 4.5.5 Residuals for GLMs . . . 13

**4.6 Fitting generalized linear models** **14**
- 4.6.1 Newton–Raphson method . . . 14
- 4.6.2 Fisher scoring method . . . 15
- 4.6.3 ML as iterative reweighted least squares . . . 15
- 4.6.4 Simplifications for canonical links . . . 16

GLMs contain logistic regression, Poisson loglinear models and ordinary regression.

## 4.1 Generalized linear model

**Motivation.** A GLM keeps the linear predictor but chooses the response distribution and the linearizing scale separately.

### 4.1.1 Components of generalized linear models

**Definition (Natural exponential family)**

A family of the form

$$f(y_i;\theta_i) = a(\theta_i)\, b(y_i) \exp\{y_i\, Q(\theta_i)\} \tag{4.1}$$

is a *natural exponential family*; $Q(\theta)$ is the *natural parameter*.

**Definition (Generalized linear model)**

(i) *Random component*: independent $y_1, \ldots, y_N$, each from (4.1) (more generally, the exponential dispersion family of Section 4.4).
(ii) *Systematic component*: a *linear predictor* $\eta_i = \sum_{j=1}^{p} \beta_j x_{ij}$ (usually $x_{i1} \equiv 1$, an intercept $\alpha$).
(iii) *Link function*: a monotone differentiable $g$ with

$$g(\mu_i) = \eta_i = \sum_j \beta_j x_{ij}, \qquad \mu_i = E(Y_i). \tag{4.2}$$

The link with $g(\mu_i) = Q(\theta_i)$ is the *canonical link*; $g(\mu) = \mu$ is the *identity link*.

### 4.1.2 Binomial logit models for binary data

For $Y \in \{0, 1\}$ with $P(Y = 1) = \pi$, $f(y;\pi) = (1-\pi)\exp\{y \log[\pi/(1-\pi)]\}$, which is (4.1) with $a(\pi) = 1 - \pi$, $b(y) = 1$, $Q(\pi) = \operatorname{logit} \pi$. The logit is the canonical link; GLMs with it are *logit models*.

### 4.1.3 Poisson loglinear models for count data

For a Poisson count, $f(y;\mu) = e^{-\mu}(1/y!)\exp(y \log \mu)$, so $Q(\mu) = \log \mu$ and the canonical link is the log link:

$$\log \mu_i = \sum_j \beta_j x_{ij}, \qquad i = 1, \ldots, N, \tag{4.3}$$

the *Poisson loglinear model*.

### 4.1.4 Generalized linear models for continuous responses

The normal (with dispersion parameter) has the mean as natural parameter, so regression, ANOVA and ANCOVA are identity-link GLMs.

| Random component | Link | Systematic | Model |
|---|---|---|---|
| Normal | identity | cont./categ./mixed | regression / ANOVA / ANCOVA |
| Binomial | logit | mixed | logistic regression (Ch. 5–6) |
| Poisson | log | mixed | loglinear (Ch. 8–9) |
| Multinomial | generalized logit | mixed | multinomial response (Ch. 7) |

Unlike transforming $Y$, the link need not stabilize variance or produce normality: fitting maximizes the chosen likelihood.

### 4.1.5 Deviance

Let $L(\boldsymbol{\mu}; \mathbf{y})$ be the log likelihood in terms of the means. The *saturated model* ($\hat{\mu}_i = y_i$, one parameter per observation) attains the largest value $L(\mathbf{y}; \mathbf{y})$.

**Definition (Deviance)**

For a Poisson or binomial GLM, $D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = -2[L(\hat{\boldsymbol{\mu}}; \mathbf{y}) - L(\mathbf{y}; \mathbf{y})]$, the LR statistic for the model against the saturated alternative.

With $N$ fixed and growing counts it is asymptotically $\chi^2_{N-p}$ (Theorem 4.19).

**Proposition 4.1 (Deviance of the homogeneity model)**

Let $y_i \sim \text{bin}(n_i, \pi_i)$ independently, $i = 1, \ldots, N$, and consider the model $\pi_i = \pi$ for all $i$ ($p = 1$). Its deviance equals the statistic $G^2$ for testing independence in the $N \times 2$ table of successes and failures, and has df $= N - 1$.

**Proof.**

*Step 1.* Let $n = \sum n_i$, $s = \sum y_i$. With $L = \sum_i [y_i \log \pi_i + (n_i - y_i) \log(1 - \pi_i)]$, the MLEs are $\hat{\pi} = s/n$ (model) and $\tilde{\pi}_i = y_i/n_i$ (saturated).

*Step 2.* Hence

$$D = 2 \sum_i \left[ y_i \log \frac{y_i}{n_i s/n} + (n_i - y_i) \log \frac{n_i - y_i}{n_i(n-s)/n} \right].$$

*Step 3.* The independence fitted values of the $N \times 2$ table are $n_i s/n$ and $n_i(n-s)/n$, so $D = 2 \sum \text{obs} \log(\text{obs}/\text{fit}) = G^2$, with df $= N - 1$ ($N$ saturated parameters minus one). ■

### 4.1.6 Advantages of the GLM formulation

One theory and one algorithm (IRLS, Section 4.6) cover every exponential-family response and link; deviances generalize ANOVA (Section 4.5).

## 4.2 Generalized linear models for binary data

**Motivation.** For binary $Y$, the choice of which function of $\pi(x)$ is linear in $x$ determines how effects are interpreted.

### 4.2.1 Linear probability model

The *linear probability model* $\pi(x) = \alpha + \beta x$ is the identity-link binomial GLM; $\beta$ is the change in probability per unit $x$, but $\alpha + \beta x$ eventually leaves $[0, 1]$. Fit by binomial ML, not least squares (the variance depends on $x$); they agree only for fitted values in about 0.3–0.7.

### 4.2.2 Snoring and heart disease example

**Example 4.2 (Snoring and heart disease)**

2484 subjects, snoring scores $x = (0, 2, 4, 5)$, independent binomial rows.

| Snoring | Yes | No | Proportion | Linear fit | Logit fit |
|---|---|---|---|---|---|
| Never ($x = 0$) | 24 | 1355 | 0.017 | 0.017 | 0.021 |
| Occasionally ($x = 2$) | 35 | 603 | 0.055 | 0.057 | 0.044 |
| Nearly every night ($x = 4$) | 21 | 192 | 0.099 | 0.096 | 0.093 |
| Every night ($x = 5$) | 30 | 224 | 0.118 | 0.116 | 0.132 |

ML linear fit $\hat{\pi}(x) = 0.0172 + 0.0198x$ ($\widehat{\text{se}} = 0.0028$): risk 0.02 for nonsnorers, +0.04 per two score units. Logistic fit $\text{logit}\,\hat{\pi}(x) = -3.87 + 0.40x$; over this range the fits are similar (Figure 1).

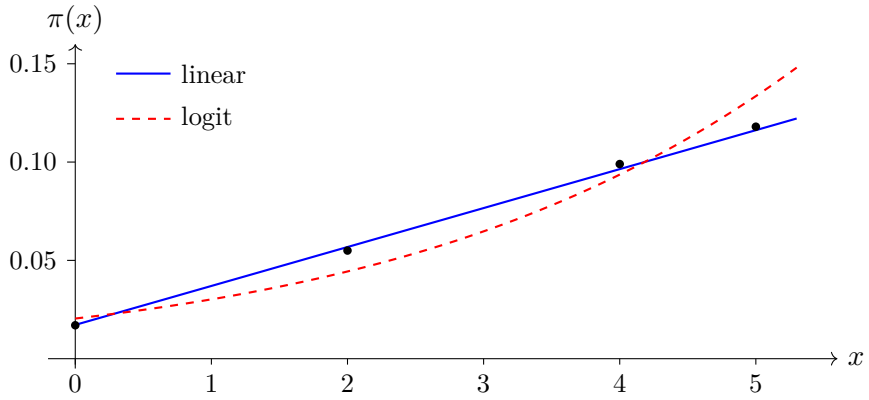



Figure 1: Snoring data: sample proportions with the linear probability and logistic fits.

### 4.2.3 Logistic regression model

Usually a fixed change in $x$ matters less when $\pi(x)$ is near 0 or 1, giving S-shaped curves such as

$$\pi(x) = \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)}. \tag{4.4}$$

**Proposition 4.3 (Logistic regression is the binomial GLM with logit link)**

Model (4.4) holds if and only if

$$\log \frac{\pi(x)}{1 - \pi(x)} = \alpha + \beta x.$$

Hence logistic regression is the binomial GLM with the canonical (logit) link.

**Proof.** From (4.4), $1-\pi(x) = 1/[1+\exp(\alpha+\beta x)]$, so the odds are $\exp(\alpha+\beta x)$; take logs. Conversely, solving $\log[\pi/(1-\pi)] = \eta$ gives $\pi = e^\eta/(1+e^\eta)$. The logit is the binomial natural parameter (Section 4.1.2). ■

The logit ranges over $\mathbb{R}$, removing the structural problem; for the snoring data $\hat\beta = 0.40 > 0$.

### 4.2.4 Binomial GLM for $2\times 2$ contingency tables

**Proposition 4.4 (Effect parameters for a binary predictor)**

Let $x \in \{0,1\}$ and $\text{link}[\pi(x)] = \alpha+\beta x$. Then $\beta = \text{link}[\pi(1)] - \text{link}[\pi(0)]$; in particular
(a) identity link: $\beta = \pi(1)-\pi(0)$, the difference of proportions;
(b) log link: $\beta = \log[\pi(1)/\pi(0)]$, the log relative risk;
(c) logit link: $\beta = \log\dfrac{\pi(1)/[1-\pi(1)]}{\pi(0)/[1-\pi(0)]}$, the log odds ratio.

**Proof.** Subtract $\text{link}[\pi(0)] = \alpha$ from $\text{link}[\pi(1)] = \alpha+\beta$ and specialize the link; for (c), a difference of log odds is a log odds ratio. ■

### 4.2.5 Probit and inverse cdf link functions

A monotone curve has the shape of a cdf: for a standard cdf $\Phi$ of a location–scale class,

$$\pi(x) = \Phi(\alpha+\beta x) \quad\Longleftrightarrow\quad \Phi^{-1}[\pi(x)] = \alpha+\beta x, \tag{4.5}$$

with link $\Phi^{-1}$. $\Phi$ standard normal gives the *probit model* (Section 6.6); the standard logistic cdf $e^x/(1+e^x)$ (sd $\pi/\sqrt{3}$, slightly heavier tails) has inverse logit $\pi$.

## 4.3 Generalized linear models for counts

**Motivation.** Counts have variance growing with the mean: the Poisson log-linear GLM is basic, overdispersion leads to the negative binomial.

### 4.3.1 Poisson loglinear models

With one predictor,

$$\log\mu = \alpha+\beta x, \qquad \mu = \exp(\alpha+\beta x) = e^\alpha(e^\beta)^x, \tag{4.6}$$

so a one-unit increase in $x$ multiplies the mean by $e^\beta$.

### 4.3.2 Horseshoe crab mating example

**Example 4.5 (Horseshoe crab satellites)**

173 female crabs; $Y$ = number of satellite males; predictors color, spine condition, weight, width $x$ (mean 26.3 cm). Grouped by width:

| Width (cm) | $<23.25$ | $-24.25$ | $-25.25$ | $-26.25$ | $-27.25$ | $-28.25$ | $-29.25$ | $>29.25$ |
|---|---|---|---|---|---|---|---|---|
| Cases | 14 | 14 | 28 | 39 | 22 | 24 | 18 | 14 |
| Mean | 1.00 | 1.43 | 2.39 | 2.69 | 2.86 | 3.87 | 3.94 | 5.14 |
| Variance | 2.77 | 8.88 | 6.54 | 11.38 | 6.88 | 8.81 | 16.88 | 8.29 |

(a) Log link: $\log\hat{\mu} = -3.305 + 0.164x$ ($\widehat{\text{se}} = 0.020$); $\hat{\mu} = 2.74$ at $x = 26.3$, and $e^{0.164} = 1.18$: each cm raises the mean by 18%.
(b) Identity link: $\hat{\mu} = -11.53 + 0.55x$ ($\widehat{\text{se}} = 0.06$). The fits agree where most data lie.

### 4.3.3 Overdispersion for Poisson GLMs

Above, the variances far exceed the means: *overdispersion*, typically from unobserved heterogeneity.

**Proposition 4.6 (Mixtures of Poissons are overdispersed)**

If $Y \mid \lambda \sim \text{Poisson}(\lambda)$ and $\lambda$ is random with $E(\lambda) = \mu$, then $E(Y) = \mu$ and $\text{Var}(Y) = \mu + \text{Var}(\lambda) \geq \mu$, with equality only when $\lambda$ is degenerate.

**Proof.** [Supplement] $E(Y) = E[E(Y \mid \lambda)] = \mu$ and $\text{Var}(Y) = E[\text{Var}(Y \mid \lambda)] + \text{Var}[E(Y \mid \lambda)] = \mu + \text{Var}(\lambda)$. ■

If the mean is right but the distribution is not Poisson, the ML estimates stay consistent but the standard errors are wrong.

### 4.3.4 Negative binomial GLMs

The negative binomial mass function is

$$f(y; k, \mu) = \frac{\Gamma(y+k)}{\Gamma(k)\Gamma(y+1)}\Big(\frac{k}{\mu+k}\Big)^k\Big(1-\frac{k}{\mu+k}\Big)^y, \qquad y = 0, 1, 2, \ldots \tag{4.7}$$

**Proposition 4.7 (Negative binomial as a gamma mixture)**

If $Y \mid \lambda \sim \text{Poisson}(\lambda)$ and $\lambda$ has the gamma density with shape $k$ and mean $\mu$, then $Y$ has the mass function (4.7), with $E(Y) = \mu$ and $\text{Var}(Y) = \mu + \mu^2/k$. For fixed $k$, (4.7) is a natural exponential family with natural parameter $\log[\mu/(\mu+k)]$, so negative binomial regression is a GLM; as $k^{-1} \to 0$ it tends to the Poisson.

**Proof.** [Supplement]

*Step 1.* With $h(\lambda) = (k/\mu)^k\lambda^{k-1}e^{-k\lambda/\mu}/\Gamma(k)$,

$$P(Y=y) = \frac{(k/\mu)^k}{\Gamma(k)\,y!}\int_0^\infty \lambda^{y+k-1}e^{-\lambda(1+k/\mu)}d\lambda = \frac{\Gamma(y+k)}{\Gamma(k)\,y!}\Big(\frac{k}{\mu+k}\Big)^k\Big(\frac{\mu}{\mu+k}\Big)^y,$$

which is (4.7).

*Step 2.* The gamma law has mean $\mu$, variance $\mu^2/k$; Proposition 4.6 gives the moments.

*Step 3.* With $Q = \log[\mu/(\mu+k)]$, (4.7) is $(1-e^Q)^k\,b(y)\exp(yQ)$, $b(y) = \Gamma(y+k)/[\Gamma(k)y!]$: the form (4.1).

*Step 4.* As $k \to \infty$, $\Gamma(y+k)/[\Gamma(k)(\mu+k)^y] \to 1$ and $(1+\mu/k)^{-k} \to e^{-\mu}$, so $f(y) \to e^{-\mu}\mu^y/y!$. ■

$k^{-1}$ is a *dispersion parameter* (Section 13.4).

**Example 4.8 (Crab satellites: negative binomial fit)**

Identity link: Poisson $\hat{\mu} = -11.53 + 0.55x$ ($\widehat{\text{se}} = 0.06$); negative binomial $\hat{\mu} = -11.15 + 0.53x$ ($\widehat{\text{se}} = 0.11$), $\hat{k}^{-1} = 0.98$. Similar fits, but the honest SE is nearly doubled.

### 4.3.5 Poisson regression for rates

For a rate $\mu_i/t_i$ with exposure $t_i$, $\log(\mu_i/t_i) = \sum_j \beta_j x_{ij}$ is the Poisson GLM $\log \mu_i = \log t_i + \sum_j \beta_j x_{ij}$ with *offset* $\log t_i$ (Section 9.7).

### 4.3.6 Poisson GLM of independence in $I \times J$ contingency tables

**Proposition 4.9 (Poisson form of independence)**

Let $Y_{ij}$ be independent Poisson with means $\mu_{ij} = \mu\alpha_i\beta_j$, where $\alpha_i, \beta_j > 0$ and $\sum_i \alpha_i = \sum_j \beta_j = 1$; equivalently

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y, \qquad \lambda = \log \mu,\ \lambda_i^X = \log \alpha_i,\ \lambda_j^Y = \log \beta_j. \tag{4.8}$$

Conditional on $n = \sum_{i,j} Y_{ij}$, the cell counts are multinomial with probabilities $\pi_{ij} = \alpha_i\beta_j$, which satisfy $\pi_{i+} = \alpha_i$, $\pi_{+j} = \beta_j$ and $\pi_{ij} = \pi_{i+}\pi_{+j}$.

**Proof.**

*Step 1.* $n \sim \text{Poisson}(\sum \mu_{ij}) = \text{Poisson}(\mu)$.

*Step 2.* For counts $\{n_{ij}\}$ with total $n$,

$$P(\{Y_{ij} = n_{ij}\} \mid n) = \frac{\prod_{ij} e^{-\mu_{ij}} \mu_{ij}^{n_{ij}}/n_{ij}!}{e^{-\mu}\mu^n/n!} = \frac{n!}{\prod_{ij} n_{ij}!} \prod_{ij} \Big(\frac{\mu_{ij}}{\mu}\Big)^{n_{ij}},$$

multinomial with $\pi_{ij} = \alpha_i\beta_j$.

*Step 3.* $\pi_{i+} = \alpha_i \sum_j \beta_j = \alpha_i$ and $\pi_{+j} = \beta_j$, so $\pi_{ij} = \pi_{i+}\pi_{+j}$. ∎

So Chapter 3 independence inferences are inferences for (4.8); Chapters 8–9 extend them.

## 4.4 Moments and likelihood for generalized linear models

**Motivation.** Mean, variance, likelihood equations and information all come from derivatives of $b(\theta)$.

**Definition (Exponential dispersion family)**

Independent $y_1, \ldots, y_N$ with

$$f(y_i; \theta_i, \phi) = \exp\Big\{\frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)\Big\}; \tag{4.9}$$

$\theta_i$ is the *natural* and $\phi$ the *dispersion parameter*; usually $a(\phi) = \phi/\omega_i$ with known weight $\omega_i$. For known $\phi$ this is (4.1) with $Q(\theta) = \theta/a(\phi)$.

### 4.4.1 Mean and variance functions for the random component

With $L_i = \log f(y_i; \theta_i, \phi)$ and $L = \sum_i L_i$,

$$L_i = \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi). \tag{4.10}$$

**Theorem 4.10 (Moments of the exponential dispersion family)**

For (4.9), with $\theta_i$ interior to the natural parameter space,

$$\mu_i = E(Y_i) = b'(\theta_i), \qquad \mathrm{Var}(Y_i) = b''(\theta_i)\, a(\phi). \tag{4.11}$$

**Proof.** [Supplement]

*Step 1.* (Bartlett identities.) Let $\ell = \log f$. For $\theta$ interior, differentiation under $\int f\, dy = 1$ is permitted (support free of $\theta$, locally dominated derivatives), giving $E(\partial\ell/\partial\theta) = 0$; differentiating again with $\partial f/\partial\theta = f\, \partial\ell/\partial\theta$ gives $-E(\partial^2\ell/\partial\theta^2) = E(\partial\ell/\partial\theta)^2$ (sums for discrete $Y$).

*Step 2.* From (4.10), $\partial L_i/\partial\theta_i = [y_i - b'(\theta_i)]/a(\phi)$ and $\partial^2 L_i/\partial\theta_i^2 = -b''(\theta_i)/a(\phi)$.

*Step 3.* The first identity gives $\mu_i = b'(\theta_i)$; the second gives $b''(\theta_i)/a(\phi) = \mathrm{Var}(Y_i)/a(\phi)^2$, i.e. $\mathrm{Var}(Y_i) = b''(\theta_i)a(\phi)$. ■

Since $b'' > 0$, $b'$ is strictly increasing, so $\theta_i$ and $\mu_i$ determine each other.

### 4.4.2 Mean and variance functions for Poisson and binomial

**Proposition 4.11 (Poisson and binomial in dispersion form)**

(a) Poisson: $\theta_i = \log\mu_i$, $b(\theta_i) = e^{\theta_i}$, $a(\phi) = 1$, $c = -\log y_i!$; then $E(Y_i) = \mathrm{Var}(Y_i) = \mu_i$.
(b) Binomial proportion: if $n_i Y_i \sim \mathrm{bin}(n_i, \pi_i)$, then $\theta_i = \log[\pi_i/(1-\pi_i)]$, $b(\theta_i) = \log(1 + e^{\theta_i})$, $a(\phi) = 1/n_i$, $c = \log\binom{n_i}{n_i y_i}$; then $E(Y_i) = \pi_i$ and $\mathrm{Var}(Y_i) = \pi_i(1-\pi_i)/n_i$.

**Proof.**

(a) $e^{-\mu_i}\mu_i^{y_i}/y_i! = \exp[y_i\theta_i - e^{\theta_i} - \log y_i!]$, and $b'(\theta_i) = b''(\theta_i) = e^{\theta_i} = \mu_i$ (Theorem 4.10).

(b) Since $\log(1-\pi_i) = -\log(1+e^{\theta_i})$,

$$\binom{n_i}{n_i y_i}\pi_i^{n_i y_i}(1-\pi_i)^{n_i - n_i y_i} = \exp\Big[\frac{y_i\theta_i - \log(1+e^{\theta_i})}{1/n_i} + \log\binom{n_i}{n_i y_i}\Big].$$

Then $b'(\theta_i) = e^{\theta_i}/(1+e^{\theta_i}) = \pi_i$ and $b''(\theta_i)a(\phi) = \pi_i(1-\pi_i)/n_i$. ■

For $N(\mu_i, \sigma^2)$: $\theta_i = \mu_i$, $b(\theta) = \theta^2/2$, $a(\phi) = \sigma^2$.

### 4.4.3 Systematic component and link function

In matrix form $\boldsymbol{\eta} = X\boldsymbol{\beta}$ with $N \times p$ *model matrix* $X$, and $\eta_i = g(\mu_i)$.

**Proposition 4.12 (Canonical link)**

The canonical link ($g(\mu_i) = \theta_i$) is $g = (b')^{-1}$: log for the Poisson, logit for the binomial.

**Proof.** By Theorem 4.10, $\mu_i = b'(\theta_i)$ with $b'$ strictly increasing, so $\theta_i = (b')^{-1}(\mu_i)$. Poisson: $b' = \exp$, inverse log. Binomial: $b'(\theta) = e^\theta/(1+e^\theta)$, inverse logit. ■

### 4.4.4 Likelihood equations for a GLM

The log likelihood

$$L(\boldsymbol{\beta}) = \sum_i \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + \sum_i c(y_i, \phi) \tag{4.12}$$

depends on $\boldsymbol{\beta}$ through $\theta_i(\mu_i)$, $\mu_i = g^{-1}(\eta_i)$, $\eta_i = \sum_j \beta_j x_{ij}$.

**Theorem 4.13 (GLM likelihood equations)**

For a GLM with exponential dispersion random component,

$$\frac{\partial L_i}{\partial \beta_j} = \frac{(y_i - \mu_i)x_{ij}}{\mathrm{Var}(Y_i)} \frac{\partial \mu_i}{\partial \eta_i}, \tag{4.13}$$

and the likelihood equations are

$$\sum_{i=1}^{N} \frac{(y_i - \mu_i)x_{ij}}{\mathrm{Var}(Y_i)} \frac{\partial \mu_i}{\partial \eta_i} = 0, \qquad j = 1, \ldots, p. \tag{4.14}$$

**Proof.**

*Step 1.* By the chain rule, $\partial L_i/\partial \beta_j = (\partial L_i/\partial \theta_i)(\partial \theta_i/\partial \mu_i)(\partial \mu_i/\partial \eta_i)(\partial \eta_i/\partial \beta_j)$.

*Step 2.* $\partial L_i/\partial \theta_i = (y_i - \mu_i)/a(\phi)$ (proof of Theorem 4.10); $\partial \mu_i/\partial \theta_i = b''(\theta_i) = \mathrm{Var}(Y_i)/a(\phi)$; $\partial \eta_i/\partial \beta_j = x_{ij}$.

*Step 3.* Multiplying, $a(\phi)$ cancels, giving (4.13); summing over $i$ gives (4.14). ■

The equations depend on the distribution only through $\mu_i$ and the *variance function* $v(\mu_i) = \mathrm{Var}(Y_i)$: $\mu$ (Poisson), $\mu(1-\mu)$ (Bernoulli), $\sigma^2$ (normal).

### 4.4.5 Likelihood equations for binomial GLMs

**Corollary 4.14 (Binomial likelihood equations)**

Let $n_i Y_i \sim \mathrm{bin}(n_i, \pi_i)$ and $\pi_i = \Phi(\eta_i)$, $\eta_i = \sum_j \beta_j x_{ij}$, for a continuous cdf $\Phi$ with density $\phi$. The likelihood equations are

$$\sum_i \frac{n_i(y_i - \pi_i)\, x_{ij}}{\pi_i(1-\pi_i)} \phi(\eta_i) = 0, \qquad j = 1, \ldots, p. \tag{4.15}$$

For the logit link they reduce to

$$\sum_i n_i(y_i - \pi_i)x_{ij} = 0, \qquad j = 1, \ldots, p. \tag{4.16}$$

**Proof.** Substitute $\partial \pi_i/\partial \eta_i = \phi(\eta_i)$ and $\mathrm{Var}(Y_i) = \pi_i(1-\pi_i)/n_i$ in (4.14). For the logit, $\partial \eta_i/\partial \pi_i = 1/[\pi_i(1-\pi_i)]$, so $\phi(\eta_i) = \pi_i(1-\pi_i)$ cancels the denominator. ■

### 4.4.6 Asymptotic covariance matrix of model parameter estimators

**Theorem 4.15 (Information matrix of a GLM)**

The expected information matrix $\mathcal{I}$ with entries $E[-\partial^2 L(\boldsymbol{\beta})/\partial \beta_h \partial \beta_j]$ is

$$\mathcal{I} = X^\top W X, \qquad W = \mathrm{diag}(w_1, \ldots, w_N), \quad w_i = \frac{(\partial \mu_i/\partial \eta_i)^2}{\mathrm{Var}(Y_i)}. \tag{4.17}$$

**Proof.**

*Step 1.* By the second Bartlett identity applied to $\boldsymbol{\beta}$, $E(\partial^2 L_i/\partial \beta_h \partial \beta_j) = -E(\partial L_i/\partial \beta_h \cdot \partial L_i/\partial \beta_j)$.

*Step 2.* By (4.13) and $E(Y_i-\mu_i)^2 = \mathrm{Var}(Y_i)$, the latter equals $x_{ih}x_{ij}(\partial\mu_i/\partial\eta_i)^2/\mathrm{Var}(Y_i) = x_{ih}w_ix_{ij}$.

*Step 3.* Summing over independent observations gives the $(h,j)$ entry of $X^\top WX$. ■

Hence

$$\widehat{\mathrm{Cov}}(\hat{\boldsymbol{\beta}}) = \hat{\mathcal{I}}^{-1} = (X^\top \hat{W} X)^{-1}, \tag{4.18}$$

justified by the next theorem.

**Theorem 4.16 (Asymptotic normality of the ML estimator)**

Let $\boldsymbol{\beta}_0$ be the true value, interior to the parameter space, and write $\boldsymbol{u}(\boldsymbol{\beta}) = \partial L/\partial\boldsymbol{\beta}$, $H(\boldsymbol{\beta}) = \partial^2 L/\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^\top$, $\mathcal{I}_N(\boldsymbol{\beta}) = X^\top W(\boldsymbol{\beta})X$. Assume:
(R1) $N^{-1}\mathcal{I}_N(\boldsymbol{\beta}_0) \to \Sigma$, positive definite;
(R2) Lindeberg condition: for each $\boldsymbol{c} \in \mathbb{R}^p$, the terms $\boldsymbol{c}^\top \partial L_i/\partial\boldsymbol{\beta}$ $(i \le N)$, normalized by $(N\boldsymbol{c}^\top\Sigma\boldsymbol{c})^{1/2}$, satisfy Lindeberg's condition;
(R3) $\sup_{\|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\le\delta} \|N^{-1}H(\boldsymbol{\beta}) + \Sigma\| \xrightarrow{p} 0$ as $N\to\infty$ and then $\delta \to 0$.
Then with probability tending to one the likelihood equations have a root $\hat{\boldsymbol{\beta}}$ with $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$, and

$$\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N_p(\mathbf{0}, \Sigma^{-1}), \qquad \text{i.e.} \qquad \hat{\boldsymbol{\beta}} \approx N_p(\boldsymbol{\beta}_0, (X^\top W X)^{-1}).$$

**Proof.** [Supplement]

*Step 1.* (Score CLT.) The summands $\partial L_i/\partial\boldsymbol{\beta}$ are independent, mean $\mathbf{0}$, with covariances summing to $\mathcal{I}_N(\boldsymbol{\beta}_0)$. By (R1), (R2), Lindeberg–Feller and Cramér–Wold, $N^{-1/2}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{d} N_p(\mathbf{0},\Sigma)$; so $N^{-1}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{p} \mathbf{0}$.

*Step 2.* (Consistent root.) Let $\lambda > 0$ be the smallest eigenvalue of $\Sigma$, and fix small $\delta$ so that, by (R3), with probability $\to 1$ all eigenvalues of $-N^{-1}H$ are $\ge \lambda/2$ on $B_\delta = \{\|\boldsymbol{\beta} - \boldsymbol{\beta}_0\| \le \delta\}$. For $\boldsymbol{\beta} \in \partial B_\delta$, Taylor's theorem gives

$$L(\boldsymbol{\beta}) - L(\boldsymbol{\beta}_0) = \boldsymbol{u}(\boldsymbol{\beta}_0)^\top(\boldsymbol{\beta}-\boldsymbol{\beta}_0) + \tfrac{1}{2}(\boldsymbol{\beta}-\boldsymbol{\beta}_0)^\top H(\bar{\boldsymbol{\beta}})(\boldsymbol{\beta}-\boldsymbol{\beta}_0) \le N\delta\Big(\frac{\|\boldsymbol{u}(\boldsymbol{\beta}_0)\|}{N} - \frac{\lambda\delta}{4}\Big),$$

negative with probability $\to 1$ by Step 1. So the maximum of $L$ on the compact $B_\delta$ is interior, where $\boldsymbol{u} = \mathbf{0}$; as $\delta$ is arbitrary, $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$.

*Step 3.* (Expansion.) Coordinatewise mean value theorem: $\mathbf{0} = \boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \boldsymbol{u}(\boldsymbol{\beta}_0) + \tilde{H}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)$, rows of $\tilde{H}$ evaluated at intermediate points; by consistency and (R3), $-N^{-1}\tilde{H} \xrightarrow{p} \Sigma$.

*Step 4.* So $\sqrt{N}(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}_0) = (-N^{-1}\tilde{H})^{-1}N^{-1/2}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{d} N_p(\mathbf{0},\Sigma^{-1})$ by Slutsky; and $N^{-1}X^\top\hat{W}X \xrightarrow{p} \Sigma$ by continuity, justifying (4.18). ■

For grouped binomial or Poisson data the same holds with $N$ fixed and $n_i \to \infty$.

### 4.4.7 Likelihood equations and covariance matrix for Poisson loglinear model

**Corollary 4.17 (Poisson loglinear model)**

For $\log\boldsymbol{\mu} = X\boldsymbol{\beta}$ with Poisson responses, the likelihood equations are

$$\sum_i (y_i - \mu_i)x_{ij} = 0, \qquad j = 1,\dots,p, \tag{4.19}$$

and $\widehat{\mathrm{Cov}}(\hat{\boldsymbol{\beta}}) = (X^\top\hat{W}X)^{-1}$ with $\hat{W} = \mathrm{diag}(\hat{\mu}_1,\dots,\hat{\mu}_N)$.

**Proof.** For the log link $\partial\mu_i/\partial\eta_i = \mu_i = \mathrm{Var}(Y_i)$, so (4.14) becomes (4.19) and $w_i = \mu_i^2/\mu_i = \mu_i$. ■

## 4.5 Inference for generalized linear models

**Motivation.** Wald, score and LR methods apply to any GLM; LR inference runs through the deviance, the analogue of the residual sum of squares.

**Theorem 4.18 (Wald, likelihood-ratio and score tests)**

Under the conditions of Theorem 4.16, for $H_0 : \boldsymbol{\beta} = \boldsymbol{\beta}_0$ each of

$$\begin{aligned} W &= (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)^\top \hat{\mathcal{I}}\, (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) && \text{(Wald)}, \\ \Lambda &= -2[L(\boldsymbol{\beta}_0) - L(\hat{\boldsymbol{\beta}})] && \text{(likelihood ratio)}, \\ S &= \boldsymbol{u}(\boldsymbol{\beta}_0)^\top \mathcal{I}(\boldsymbol{\beta}_0)^{-1} \boldsymbol{u}(\boldsymbol{\beta}_0) && \text{(score)} \end{aligned}$$

converges in distribution to $\chi^2_p$, and the three differ by $o_p(1)$. For nested models $M_0 \subset M_1$, where $M_0$ imposes $q$ linear restrictions on $\boldsymbol{\beta}$, $\Lambda = -2[L(\hat{\boldsymbol{\beta}}_0) - L(\hat{\boldsymbol{\beta}}_1)] \xrightarrow{d} \chi^2_q$.

**Proof.** [Supplement] Let $\boldsymbol{Z}_N = N^{-1/2}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{d} \boldsymbol{Z} \sim N_p(\mathbf{0}, \Sigma)$; by Theorem 4.16, $\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = \Sigma^{-1}\boldsymbol{Z}_N + o_p(1)$.

*Step 1.* (Score.) $S = \boldsymbol{Z}_N^\top (N^{-1}\mathcal{I}_N)^{-1} \boldsymbol{Z}_N \xrightarrow{d} \boldsymbol{Z}^\top \Sigma^{-1} \boldsymbol{Z} \sim \chi^2_p$.

*Step 2.* (Wald.) $W = [\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)]^\top (N^{-1}\hat{\mathcal{I}})[\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)] = \boldsymbol{Z}_N^\top \Sigma^{-1} \boldsymbol{Z}_N + o_p(1)$.

*Step 3.* (LR.) Expanding about $\hat{\boldsymbol{\beta}}$, where $\boldsymbol{u} = \mathbf{0}$, $L(\boldsymbol{\beta}_0) = L(\hat{\boldsymbol{\beta}}) + \frac{1}{2}(\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}})^\top H(\bar{\boldsymbol{\beta}})(\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}})$, so by (R3) $\Lambda = \boldsymbol{Z}_N^\top \Sigma^{-1} \boldsymbol{Z}_N + o_p(1)$.

*Step 4.* (Nested.) Write $M_0$ as $\boldsymbol{\beta} = \boldsymbol{\beta}_0 + A\boldsymbol{\gamma}$, $A$ of full rank $p-q$; $M_0$ has score $A^\top \boldsymbol{u}$ and information $A^\top \mathcal{I} A$. Step 3 for both models gives

$$\Lambda = \boldsymbol{Z}_N^\top [\Sigma^{-1} - A(A^\top \Sigma A)^{-1} A^\top] \boldsymbol{Z}_N + o_p(1).$$

With $\boldsymbol{Z} = \Sigma^{1/2}\boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim N_p(\mathbf{0}, I)$, the limit is $\boldsymbol{\varepsilon}^\top (I - P)\boldsymbol{\varepsilon}$ with $P = \Sigma^{1/2} A(A^\top \Sigma A)^{-1} A^\top \Sigma^{1/2}$ a projection of rank $p - q$; so it is $\chi^2_q$. ∎

Inverting the tests gives the Wald interval $\hat{\beta}_j \pm z_{\alpha/2}\widehat{\mathrm{se}}(\hat{\beta}_j)$ and the profile-likelihood interval $\{\beta_j : \Lambda(\beta_j) \le \chi^2_1(\alpha)\}$.

### 4.5.1 Deviance and goodness of fit

Let $\tilde{\theta}_i$ ($\tilde{\mu}_i = y_i$) and $\hat{\theta}_i$ be the saturated and fitted natural parameters. By (4.12) with $a(\phi) = \phi/\omega_i$, $-2[L(\hat{\boldsymbol{\mu}}; \mathbf{y}) - L(\mathbf{y}; \mathbf{y})]$ equals the following divided by $\phi$.

**Definition (Scaled deviance and deviance)**

$$D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2 \sum_i \omega_i [y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)] \tag{4.20}$$

is the *deviance*, and $D/\phi$ the *scaled deviance*; they coincide for Poisson and binomial ($\phi = 1$). For the normal linear model $D = \sum_i (y_i - \hat{\mu}_i)^2$.

**Theorem 4.19 (Asymptotic distribution of the deviance)**

For a binomial or Poisson GLM with a fixed number $N$ of observations whose counts grow ($n_i \to \infty$, or $\mu_i \to \infty$), if the model with $p$ parameters holds then $D(\mathbf{y}; \hat{\boldsymbol{\mu}}) \xrightarrow{d} \chi^2_{N-p}$.

**Proof.** [Supplement] The saturated model is a GLM with $N$ free $\theta_i$, and the working model is the submodel given by $N - p$ linear restrictions $\boldsymbol{\theta} \in \mathrm{col}(X)$ (canonical scale; otherwise reparametrize smoothly). $D$ is the LR statistic for these nested models, so Theorem 4.18 (growing counts as sample size) gives $\chi^2_{N-p}$. ■

### 4.5.2 Deviance for Poisson models

**Proposition 4.20 (Poisson deviance)**

For a Poisson GLM,

$$D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2 \sum_i \big[ y_i \log(y_i/\hat{\mu}_i) - y_i + \hat{\mu}_i \big]. \tag{4.21}$$

If the model has log link and an intercept, $\sum_i y_i = \sum_i \hat{\mu}_i$ and $D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2 \sum_i y_i \log(y_i/\hat{\mu}_i)$.

**Proof.** Put $\hat{\theta}_i = \log \hat{\mu}_i$, $b(\hat{\theta}_i) = \hat{\mu}_i$, $\tilde{\theta}_i = \log y_i$, $b(\tilde{\theta}_i) = y_i$, $\omega_i = 1$ in (4.20). With an intercept the first equation of (4.19) is $\sum_i (y_i - \hat{\mu}_i) = 0$, so the linear terms cancel ($0 \log 0 = 0$). ■

For a two-way table under independence this is $G^2$ of Chapter 3.

### 4.5.3 Deviance for binomial models: grouped and ungrouped data

**Proposition 4.21 (Binomial deviance)**

For binomial proportions $y_i$ based on $n_i$ trials,

$$D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2 \sum_i n_i y_i \log \frac{n_i y_i}{n_i \hat{\pi}_i} + 2 \sum_i (n_i - n_i y_i) \log \frac{n_i - n_i y_i}{n_i - n_i \hat{\pi}_i}, \tag{4.22}$$

that is, $D = 2 \sum \text{observed} \times \log(\text{observed}/\text{fitted})$ over the $2N$ cells of successes and failures.

**Proof.** With $\hat{\theta}_i = \operatorname{logit} \hat{\pi}_i$, $b(\hat{\theta}_i) = -\log(1 - \hat{\pi}_i)$, $\tilde{\theta}_i = \operatorname{logit} y_i$, $b(\tilde{\theta}_i) = -\log(1 - y_i)$, $\omega_i = n_i$, (4.20) gives

$$D = 2 \sum_i n_i \Big[ y_i \log \frac{y_i}{\hat{\pi}_i} + (1 - y_i) \log \frac{1 - y_i}{1 - \hat{\pi}_i} \Big],$$

which is (4.22). ■

Grouped and ungrouped (0/1) data give the same ML estimates but different saturated models and deviances; the $\chi^2$ approximation needs grouped data with $n_i \to \infty$.

### 4.5.4 Likelihood-ratio model comparison using the deviance

**Proposition 4.22 (Deviance difference)**

Let $M_0 \subset M_1$ be nested Poisson or binomial GLMs with fitted values $\hat{\boldsymbol{\mu}}_0, \hat{\boldsymbol{\mu}}_1$. Then $D(\mathbf{y}; \hat{\boldsymbol{\mu}}_1) \leq D(\mathbf{y}; \hat{\boldsymbol{\mu}}_0)$ and the likelihood-ratio statistic for $H_0 : M_0$ given $M_1$ is

$$-2\big[L(\hat{\boldsymbol{\mu}}_0; \mathbf{y}) - L(\hat{\boldsymbol{\mu}}_1; \mathbf{y})\big] = D(\mathbf{y}; \hat{\boldsymbol{\mu}}_0) - D(\mathbf{y}; \hat{\boldsymbol{\mu}}_1),$$

asymptotically $\chi^2$ with df equal to the difference in numbers of parameters. With log link and intercept (Poisson) or for binomial data, it equals $2 \sum \text{observed} \times \log(\text{fitted}_1/\text{fitted}_0)$.

**Proof.** Maximizing over a smaller space gives $L(\hat{\boldsymbol{\mu}}_0) \leq L(\hat{\boldsymbol{\mu}}_1)$; the identity follows by adding and subtracting $L(\mathbf{y}; \mathbf{y})$; the distribution is Theorem 4.18. Subtracting (4.22) (or (4.21) without linear terms) for the two models cancels the log observed terms. ■

The saturated part cancels, so the comparison is valid for ungrouped data too.

### 4.5.5 Residuals for GLMs

**Definition (Deviance and Pearson residuals)**

With $D = \sum_i d_i$, $d_i = 2\omega_i[y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)]$, the *deviance residual* is $\sqrt{d_i}\ \operatorname{sgn}(y_i - \hat{\mu}_i)$ and the *Pearson residual* is $e_i = (y_i - \hat{\mu}_i)/[\widehat{\operatorname{Var}}(Y_i)]^{1/2}$.

For independence in a two-way table $\sum e_{ij}^2 = X^2$, $\sum d_{ij} = G^2$. These are less variable than $N(0,1)$; standardizing uses the *hat matrix*

$$\text{HatM} = W^{1/2} X (X^\top W X)^{-1} X^\top W^{1/2}. \tag{4.23}$$

**Proposition 4.23 (Leverages and residual covariance)**

(a) HatM is symmetric and idempotent with trace $p$; its diagonal elements (*leverages*) satisfy $0 \leq h_i \leq 1$ and $\sum_i h_i = p$.
(b) To first order, with $V = \operatorname{Cov}(\boldsymbol{Y}) = \operatorname{diag}\{\operatorname{Var}(Y_i)\}$,

$$\operatorname{Cov}(\boldsymbol{Y} - \hat{\boldsymbol{\mu}}) \approx V^{1/2}(I - \text{HatM})V^{1/2}, \qquad \operatorname{Var}(Y_i - \hat{\mu}_i) \approx \operatorname{Var}(Y_i)(1 - h_i).$$

Hence the *standardized Pearson residual* is

$$r_i = \frac{y_i - \hat{\mu}_i}{[\widehat{\operatorname{Var}}(Y_i)(1 - \hat{h}_i)]^{1/2}} = \frac{e_i}{\sqrt{1 - \hat{h}_i}} \qquad \left(\text{Poisson: } \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i(1 - \hat{h}_i)}}\right). \tag{4.24}$$

**Proof.** [Supplement] (a) With $A = W^{1/2}X$, $\text{HatM} = A(A^\top A)^{-1}A^\top$ is symmetric, $\text{HatM}^2 = \text{HatM}$, $\operatorname{tr}\text{HatM} = \operatorname{tr} I_p = p$, and $h_i = \sum_j h_{ij}^2 \geq h_i^2$ gives $0 \leq h_i \leq 1$.

(b)

*Step 1.* Let $G = \operatorname{diag}(\partial\eta_i/\partial\mu_i)$; by (4.13), $\boldsymbol{u} = X^\top W G(\boldsymbol{y} - \boldsymbol{\mu})$.

*Step 2.* By Theorem 4.16, $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \approx (X^\top W X)^{-1} X^\top W G(\boldsymbol{y} - \boldsymbol{\mu})$ and $\hat{\boldsymbol{\mu}} - \boldsymbol{\mu} \approx G^{-1}X(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$, so $\boldsymbol{y} - \hat{\boldsymbol{\mu}} \approx [I - G^{-1}X(X^\top W X)^{-1}X^\top W G](\boldsymbol{y} - \boldsymbol{\mu})$.

*Step 3.* Since $w_i = 1/(G_{ii}^2 \operatorname{Var}(Y_i))$, $V^{1/2} = G^{-1}W^{-1/2} =: M$, and the bracket is $M(I - \text{HatM})M^{-1}$.

*Step 4.* With $\operatorname{Cov}(\boldsymbol{y} - \boldsymbol{\mu}) = M^2$ and HatM symmetric idempotent, $\operatorname{Cov}(\boldsymbol{y} - \hat{\boldsymbol{\mu}}) \approx M(I - \text{HatM})M^{-1}M^2M^{-1}(I - \text{HatM})M = M(I - \text{HatM})M$, with $i$th diagonal entry $\operatorname{Var}(Y_i)(1 - h_i)$.

■

Leverages depend on the fit through $W$, so extreme predictor values need not have high leverage.

## 4.6 Fitting generalized linear models

**Motivation.** Newton–Raphson and Fisher scoring (that is, IRLS) solve the nonlinear likelihood equations.

### 4.6.1 Newton–Raphson method

With $\boldsymbol{u}^{(t)}, H^{(t)}$ the score and Hessian at $\boldsymbol{\beta}^{(t)}$, maximizing the quadratic approximation $L(\boldsymbol{\beta}^{(t)}) + \boldsymbol{u}^{(t)\top}(\boldsymbol{\beta}-\boldsymbol{\beta}^{(t)}) + \tfrac{1}{2}(\boldsymbol{\beta}-\boldsymbol{\beta}^{(t)})^\top H^{(t)}(\boldsymbol{\beta}-\boldsymbol{\beta}^{(t)})$ gives the *Newton–Raphson* step (Figure 2)

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - \left(H^{(t)}\right)^{-1}\boldsymbol{u}^{(t)}. \tag{4.25}$$

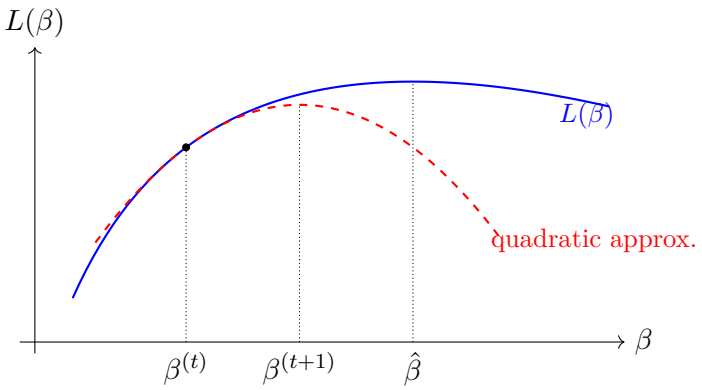



Figure 2: One Newton–Raphson cycle: the next guess maximizes the quadratic approximation at $\beta^{(t)}$.

**Example 4.24 (Newton–Raphson for a binomial parameter)**

For $y \sim \mathrm{bin}(n,\pi)$, $u = (y-n\pi)/[\pi(1-\pi)]$, $H = -[y/\pi^2 + (n-y)/(1-\pi)^2]$. From $\pi^{(0)} = \tfrac{1}{2}$ one step gives $y/n$; otherwise four or five iterations.

**Theorem 4.25 (Quadratic convergence of Newton–Raphson)**

Suppose $\boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$, $H$ is Lipschitz on a ball $B$ around $\hat{\boldsymbol{\beta}}$ with constant $\gamma$ ($\|H(\boldsymbol{a}) - H(\boldsymbol{b})\| \le \gamma\|\boldsymbol{a}-\boldsymbol{b}\|$), and $\|H(\boldsymbol{\beta})^{-1}\| \le M$ on $B$. If $\boldsymbol{\beta}^{(t)} \in B$, then

$$\|\boldsymbol{\beta}^{(t+1)} - \hat{\boldsymbol{\beta}}\| \le \frac{M\gamma}{2}\,\|\boldsymbol{\beta}^{(t)} - \hat{\boldsymbol{\beta}}\|^2.$$

In particular, if $\boldsymbol{\beta}^{(0)} \in B$ and $\|\boldsymbol{\beta}^{(0)} - \hat{\boldsymbol{\beta}}\| < 2/(M\gamma)$, the iterates converge to $\hat{\boldsymbol{\beta}}$ and the number of correct digits roughly doubles each step.

**Proof.** [Supplement]

*Step 1.* With $\boldsymbol{e}_t = \boldsymbol{\beta}^{(t)} - \hat{\boldsymbol{\beta}}$, $\boldsymbol{u}^{(t)} = \boldsymbol{u}^{(t)} - \boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \int_0^1 H(\hat{\boldsymbol{\beta}} + s\boldsymbol{e}_t)\,ds\;\boldsymbol{e}_t$.

*Step 2.* By (4.25),

$$\boldsymbol{e}_{t+1} = \boldsymbol{e}_t - (H^{(t)})^{-1}\boldsymbol{u}^{(t)} = (H^{(t)})^{-1}\int_0^1 [H(\hat{\boldsymbol{\beta}} + \boldsymbol{e}_t) - H(\hat{\boldsymbol{\beta}} + s\boldsymbol{e}_t)]ds\, \boldsymbol{e}_t.$$

*Step 3.* The integrand has norm $\leq \gamma(1-s)\|\boldsymbol{e}_t\|$ and $\int_0^1 (1-s)ds = \frac{1}{2}$, so $\|\boldsymbol{e}_{t+1}\| \leq \frac{M\gamma}{2}\|\boldsymbol{e}_t\|^2$.

*Step 4.* If $\rho = \frac{M\gamma}{2}\|\boldsymbol{e}_0\| < 1$, induction gives $\|\boldsymbol{e}_{t+1}\| \leq \rho\|\boldsymbol{e}_t\|$, the iterates stay in $B$ and $\boldsymbol{e}_t \to \mathbf{0}$, the error being roughly squared each step. ■

### 4.6.2 Fisher scoring method

*Fisher scoring* replaces $-H$ (observed information) by the expected information $\mathcal{I}$:

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + (\mathcal{I}^{(t)})^{-1}\boldsymbol{u}^{(t)} \quad \Longleftrightarrow \quad \mathcal{I}^{(t)}\boldsymbol{\beta}^{(t+1)} = \mathcal{I}^{(t)}\boldsymbol{\beta}^{(t)} + \boldsymbol{u}^{(t)}. \tag{4.26}$$

For the binomial parameter $\mathcal{I} = n/[\pi(1-\pi)]$ and one step gives $y/n$ from any start. For a GLM, $\mathcal{I}^{(t)} = X^\top W^{(t)} X$ and $\boldsymbol{u}^{(t)}$ has elements

$$u_j = \sum_{i=1}^{N} \frac{(y_i - \mu_i)x_{ij}}{\mathrm{Var}(Y_i)}\frac{\partial \mu_i}{\partial \eta_i}; \tag{4.27}$$

at convergence $(\mathcal{I}^{(t)})^{-1}$ is the covariance estimate (4.18). For canonical links the two methods coincide (Theorem 4.27); otherwise Fisher scoring has nonnegative definite $\mathcal{I}$ but need not converge quadratically.

### 4.6.3 ML as iterative reweighted least squares

For $\boldsymbol{z} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathrm{Cov}(\boldsymbol{\varepsilon}) = V$, the WLS estimator is $(X^\top V^{-1}X)^{-1}X^\top V^{-1}\boldsymbol{z}$.

**Theorem 4.26 (Fisher scoring is IRLS)**

For a GLM, the Fisher scoring step (4.26) is

$$\boldsymbol{\beta}^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} \boldsymbol{z}^{(t)},$$

the WLS fit of the *working response*

$$z_i^{(t)} = \eta_i^{(t)} + (y_i - \mu_i^{(t)})\frac{\partial \eta_i^{(t)}}{\partial \mu_i^{(t)}} \tag{4.28}$$

on $X$ with weight matrix (inverse covariance) $W^{(t)}$.

**Proof.**

*Step 1.* $\mathcal{I}^{(t)}\boldsymbol{\beta}^{(t)} = X^\top W^{(t)} X\boldsymbol{\beta}^{(t)} = X^\top W^{(t)}\boldsymbol{\eta}^{(t)}$.

*Step 2.* Since $w_i = (\partial\mu_i/\partial\eta_i)^2/\mathrm{Var}(Y_i)$, each term of (4.27) is $x_{ij}w_i(y_i - \mu_i)\,\partial\eta_i/\partial\mu_i$, so $\boldsymbol{u}^{(t)} = X^\top W^{(t)}\boldsymbol{d}^{(t)}$ with $d_i^{(t)} = (y_i - \mu_i^{(t)})\,\partial\eta_i^{(t)}/\partial\mu_i^{(t)}$.

*Step 3.* The right side of (4.26) is $X^\top W^{(t)}(\boldsymbol{\eta}^{(t)} + \boldsymbol{d}^{(t)}) = X^\top W^{(t)}\boldsymbol{z}^{(t)}$: the WLS normal equations. ■

$z_i$ linearizes the link, $g(y_i) \approx \eta_i + (y_i - \mu_i)\,\partial\eta_i/\partial\mu_i$, and $W^{-1}$ approximates its covariance. Iterating is *iteratively reweighted least squares* (IRLS); a start is $\hat{\boldsymbol{\mu}}^{(0)} = \mathbf{y}$, adjusted where $g(y_i)$ is infinite.

### 4.6.4 Simplifications for canonical links

**Theorem 4.27 (Canonical link properties)**

Consider a GLM with the canonical link, $\theta_i = \eta_i = \sum_j \beta_j x_{ij}$, and a common $a(\phi)$ for all $i$ (e.g. Poisson; binomial with all $n_i = 1$). Then:
(a) $\sum_i y_i x_{ij}$, $j = 1, \dots, p$, are sufficient statistics for $\boldsymbol{\beta}$;
(b) $\partial L_i / \partial \beta_j = (y_i - \mu_i) x_{ij} / a(\phi)$, so the likelihood equations are

$$\sum_i x_{ij} y_i = \sum_i x_{ij} \mu_i, \qquad j = 1, \dots, p, \qquad \text{i.e.} \quad X^\top \mathbf{y} = X^\top \boldsymbol{\mu}; \tag{4.29}$$

(c) $\partial^2 L / \partial \beta_h \partial \beta_j = -\sum_i x_{ih} x_{ij} b''(\theta_i)/a(\phi)$ does not involve $\mathbf{y}$; hence observed and expected information coincide, $H = -\mathcal{I}$, and Newton–Raphson equals Fisher scoring;
(d) $L$ is concave in $\boldsymbol{\beta}$, strictly if $X$ has full column rank; so a solution of (4.29), if one exists, is the unique ML estimate.
With weights $a(\phi) = \phi/\omega_i$ the same holds with $y_i, \mu_i$ replaced by $\omega_i y_i, \omega_i \mu_i$.

**Proof.**

*Step 1.* By (4.12) the only term involving both data and parameters is $a(\phi)^{-1} \sum_i y_i \theta_i = a(\phi)^{-1} \sum_j \beta_j (\sum_i y_i x_{ij})$; the factorization theorem gives (a).

*Step 2.* Here $\partial \mu_i / \partial \eta_i = b''(\theta_i) = \mathrm{Var}(Y_i)/a(\phi)$; inserting into (4.13) gives (b).

*Step 3.* Since $\partial \mu_i / \partial \beta_h = b''(\theta_i) x_{ih}$, $\partial^2 L_i / \partial \beta_h \partial \beta_j = -x_{ij} x_{ih} b''(\theta_i)/a(\phi)$, free of $y_i$ and equal to its expectation: (c).

*Step 4.* For $\boldsymbol{c} \in \mathbb{R}^p$, $\boldsymbol{c}^\top H \boldsymbol{c} = -\sum_i b''(\theta_i)(\boldsymbol{x}_i^\top \boldsymbol{c})^2/a(\phi) \le 0$, with equality only if $X\boldsymbol{c} = \mathbf{0}$, impossible for $\boldsymbol{c} \neq \mathbf{0}$ under full rank. A stationary point of a strictly concave function is its unique maximum: (d). ■

So (4.29) equates sufficient statistics to their expectations (cf. the normal equations, (4.19), (4.16) with $n_i = 1$).